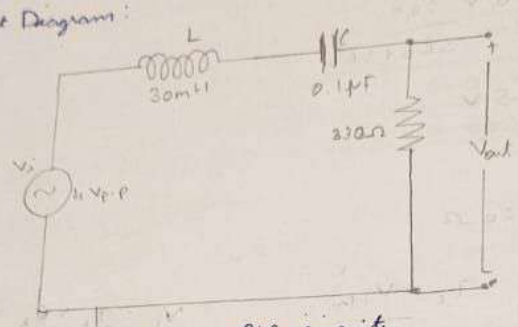
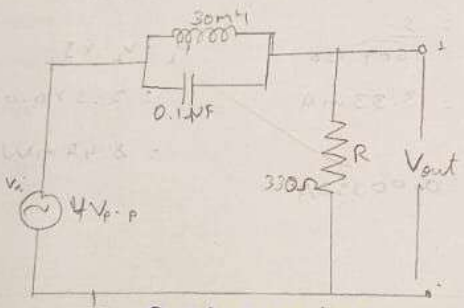
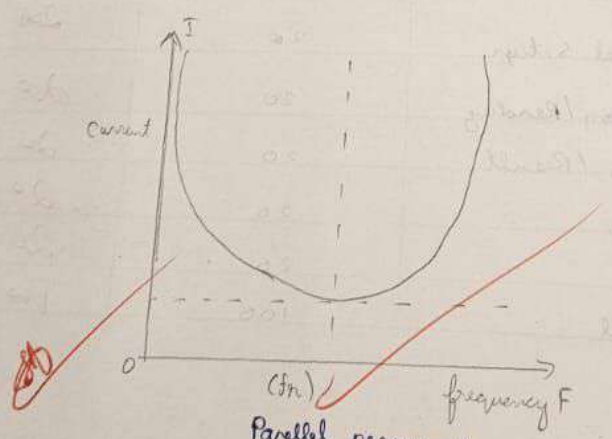
Circuit Diagram:



Series RLC circuit



Parellel RLC circuit



Parellel resonance

EXP.no: 7
Date: 10/3/26

Fre

Aim:
to st
resonance
and Q o

Apparat

S.no
01
02
03
04
05
06
07

Ex.no: 7
Date: 10/3/26

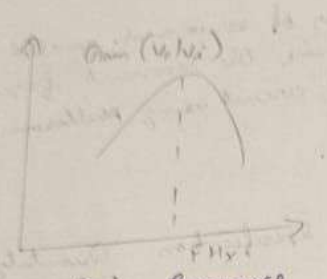
Frequency response of series and parallel resonance circuits

Aim:

to study the behaviour of series and parallel circuit at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using Multisim Simulator.

Apparatus required:

S.no	Components	Specification	Quantity
01	CRO	10MHz	1
02	DIB	-	1
03	Capacitor	0.1μF	1
04	Resistor	330Ω	1
05	Bread board	-	-
06	Connecting wires	-	-
07	PC with multisim Simulator	-	-



Series Resonance

Tabulation :

Vin: 4 Vp-p

Series Resonance

Frequency (Hz)	Output Voltage (V)	Output current (mA)
1000	1.5	4.5
2000	3.0	9.0
2900	4.0	12.1
3500	3.2	9.7
4500	2.0	6.6

Parallel Resonance

Frequency (Hz)	Output Voltag (V)	Output current (mA)
1000	3.0	3.0
2000	3.5	2.0
2900	4.0	1.2
3500	3.6	2.1
4500	3.9	3.2

O/P verified

Procedure :

1. Construct c
2. Set the inp
3. Calculate
4. Calculate
5. Repeat ste

SL.n
01
02
03
04
05

Result :

Thus
and the

Series
Resona
Bandw
Q-fact

4/3/25